



Three Illusions in Software Delivery

Joel Tosi

<https://signalsandlevers.com/3IllusionsPrDC>



Payworks



**Work
Wranglers**



CONQUEST



Google Public Sector | **carahsoft.**

improving
It's what we do.™

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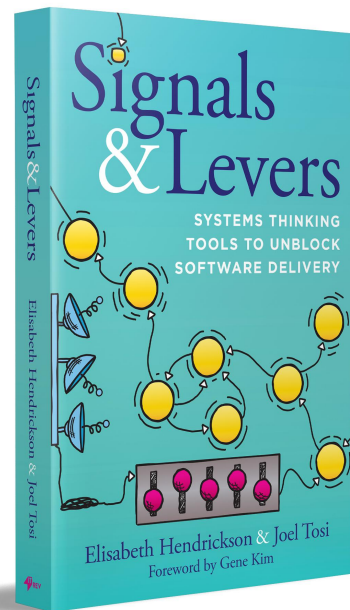
About Us



Joel Tosi



Elisabeth Hendrickson

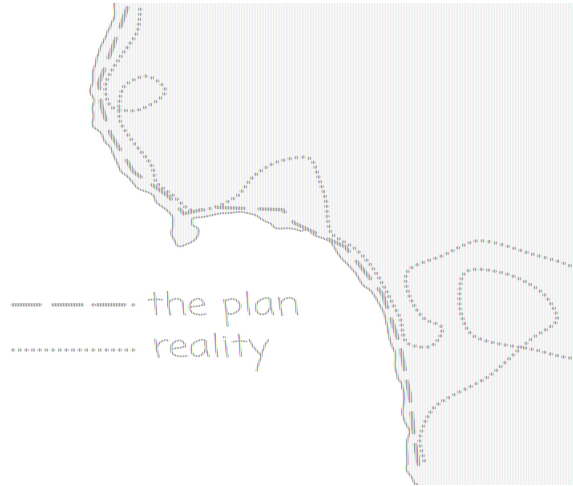


*forthcoming from
IT Revolution, Sept 2026*

Illusions...

...of Predictability

Why Are Estimates So Hard?



How long would it take to hike
the coast from San Francisco
to Newport Beach, California?

“It is in the doing of the
work that we discover the
work that we must do.

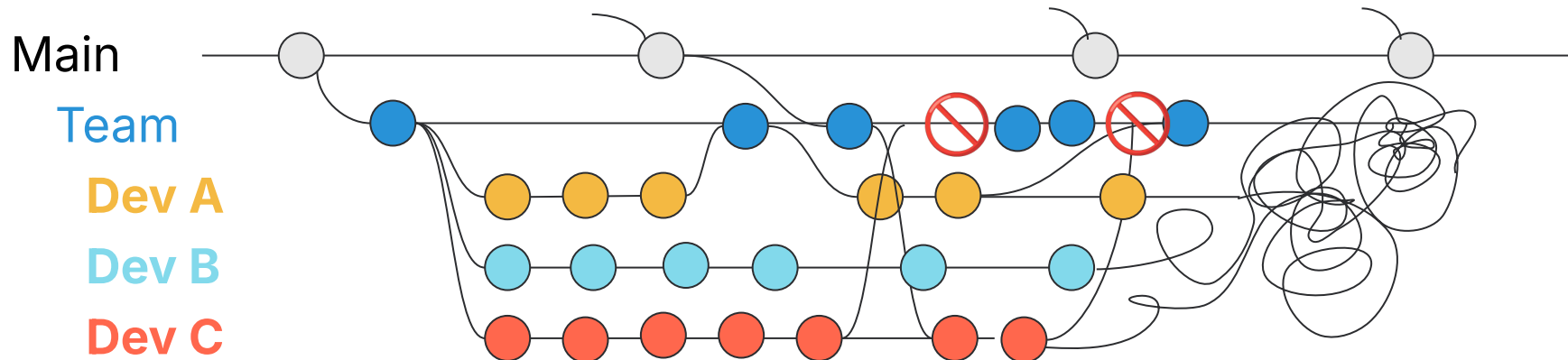
Doing exposes reality.”

– *Woody Zuill*

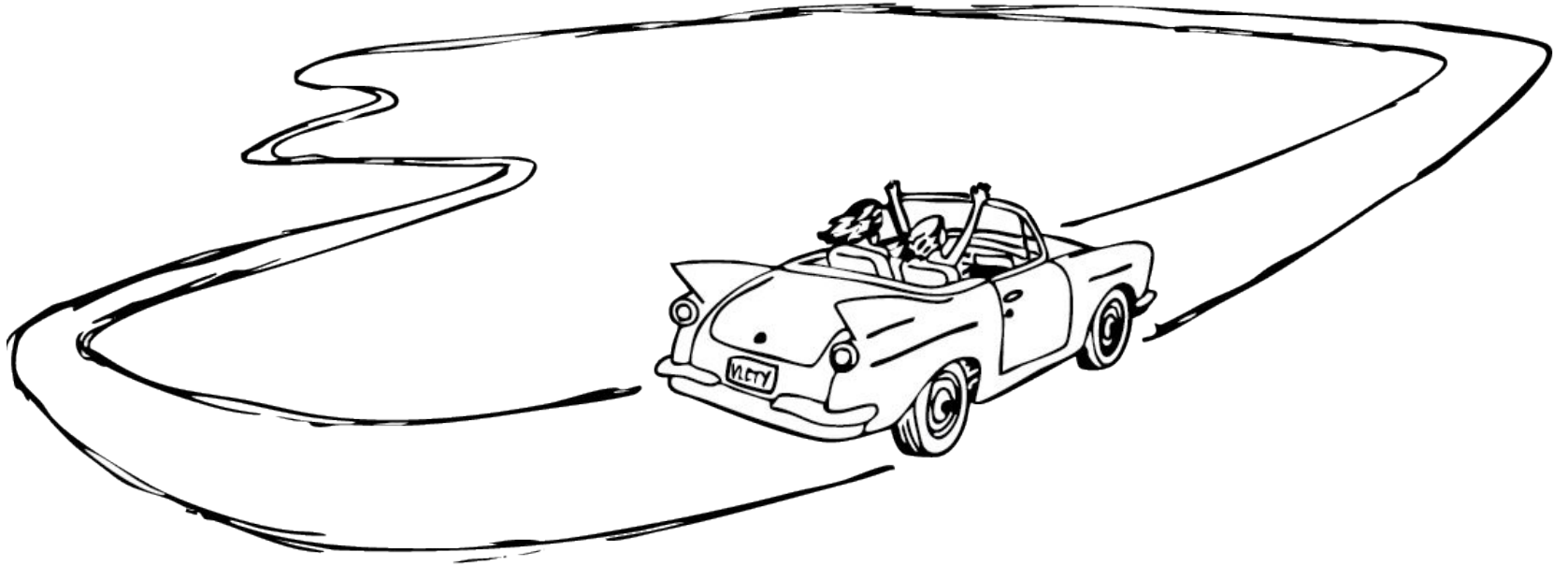


...of Progress

It Seemed Like a Good Idea at the Time...

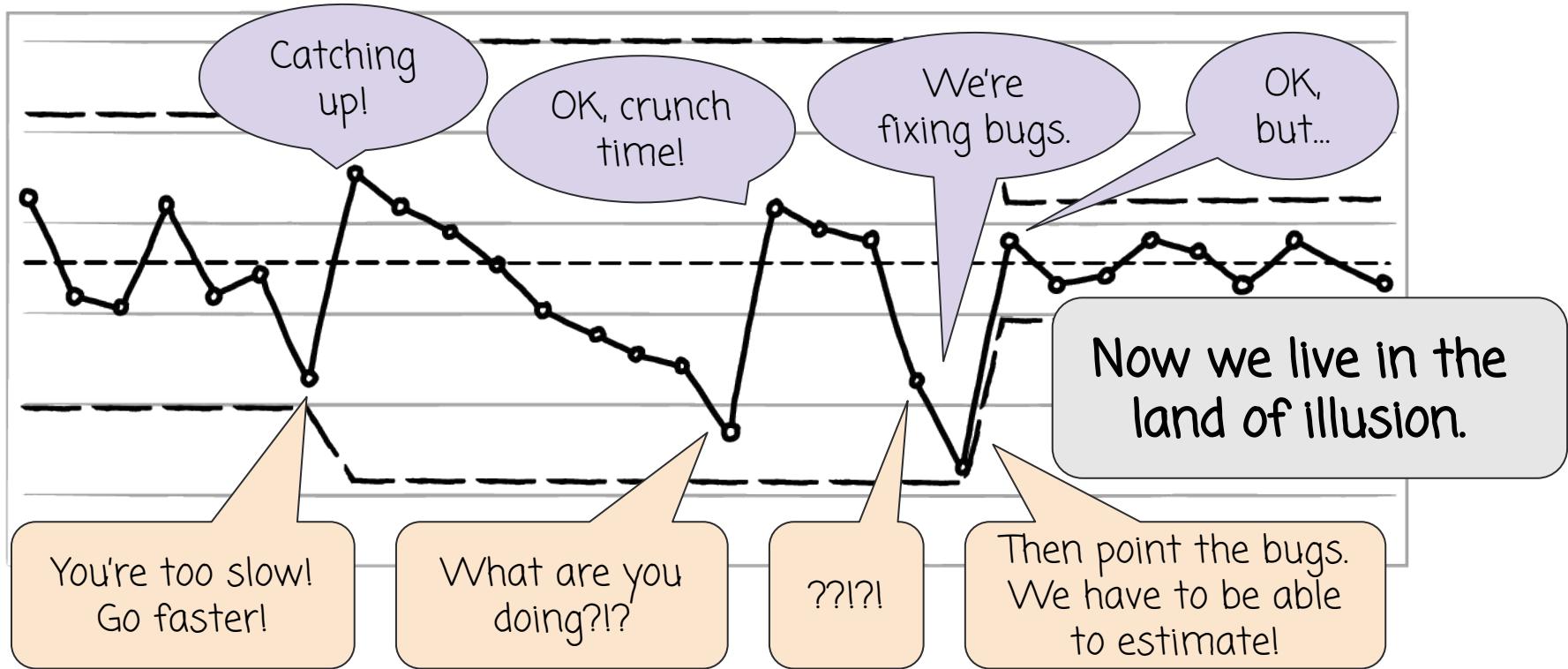


Don't Confuse Activity for Progress



...of Control

Project Pressures



It may look like a crisis,
but it's only the end of an illusion.

*Jerry Weinberg, **Secrets of Consulting***

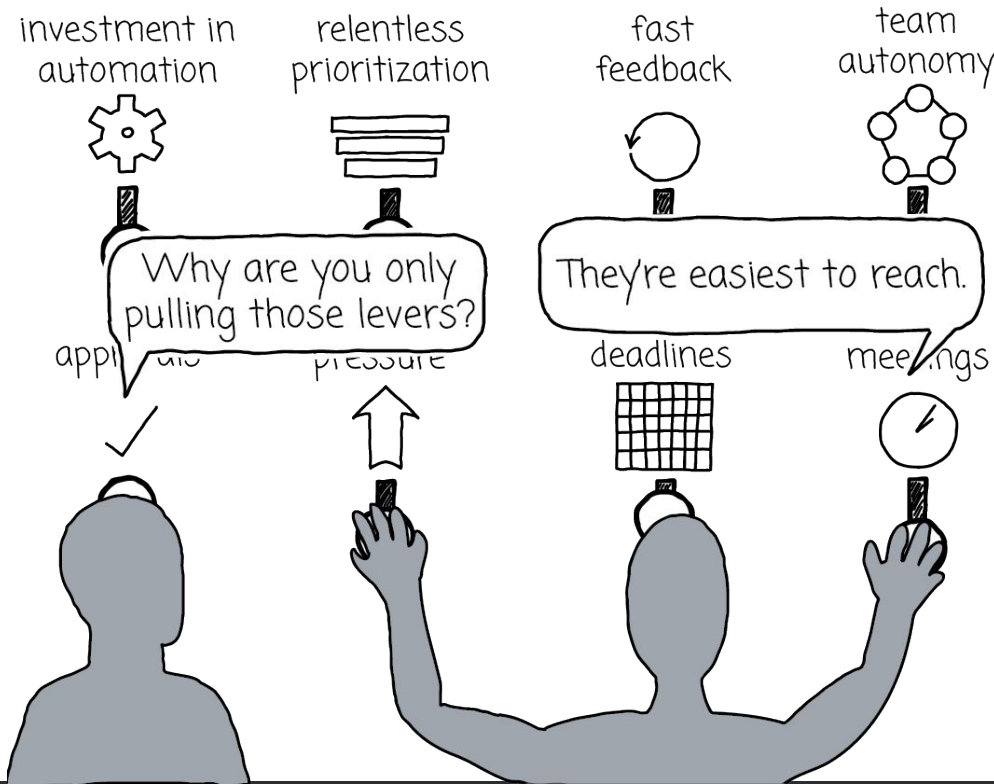
Illusions Don't Travel Alone



The illusion of progress, illusion of predictability, and illusion of control usually travel together, reinforcing each other.

Unintended Consequences

Why Best Intentions Backfire



Many of the levers that look like they'll control outcomes just shift risk around
...or, worse, hide it.

Irony Abounds in a Backfiring System

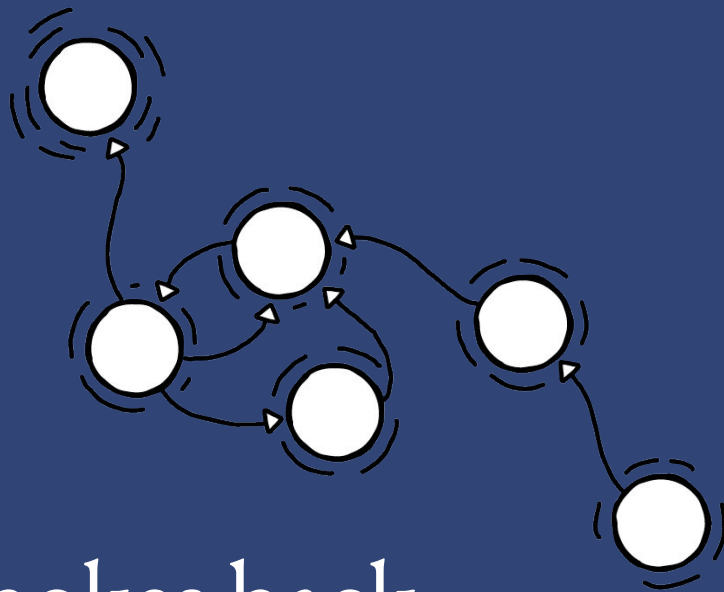
- Coordination meetings intended to keep a schedule in check reduce work time so the team has a harder time meeting the schedule.
- An open plan office was supposed to increase collaboration but everyone wears noise cancelling headphones and avoids talking.
- Meeting-free Fridays become the only day of the week when everyone is available to meet.

How to Intervene

The Rule of Systems

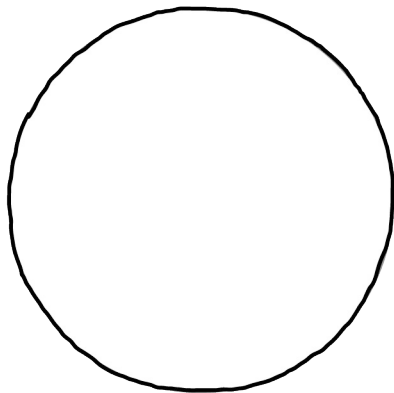
You poke the system...

...and the system pokes back.

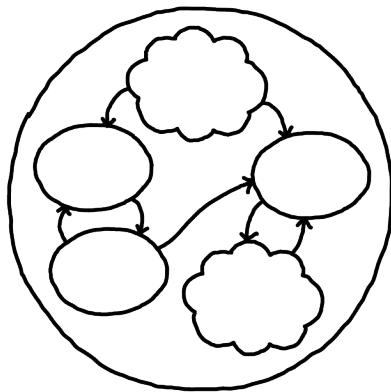


About Systems...

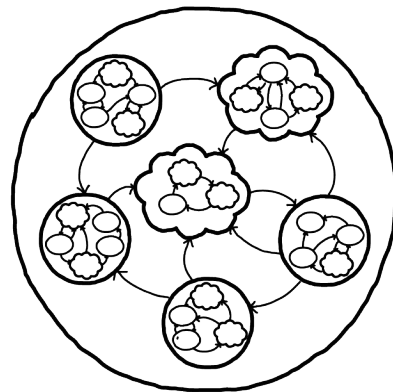
A **system** is a set of interrelated parts that does something. It has these properties:



the whole is
indivisible



parts are
interdependent



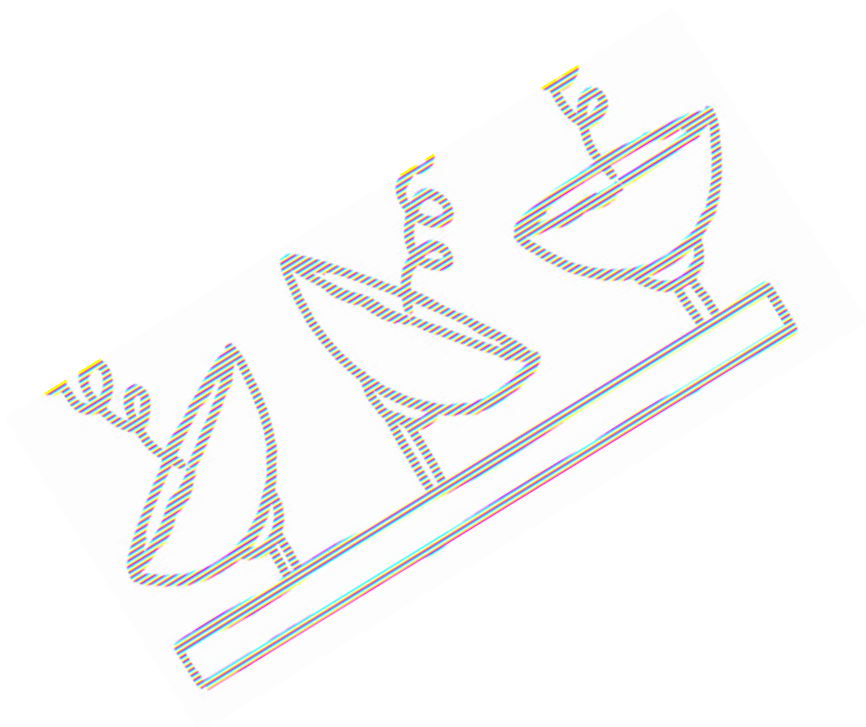
systems are
fractal

Signals

A **signal** is an observable characteristic or attribute of the system that tells you something about the way the system works.

Signals can be quantitative (lead time, cycle time, velocity, defect counts) or qualitative (quality, morale).

You are surrounded by signals.



Common Signals from Illusions (Non-Dashboard Edition)

'It's done but...'

Activity Focus

Powerpoint 'Demos'

Shipping vs Impacting

'Estimate Better'

Read The Docs

Tampering

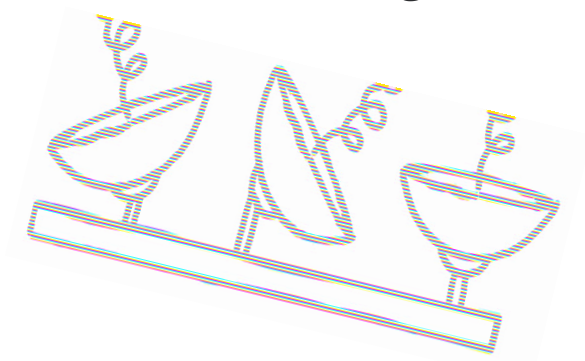
Re-defining

Meetings!

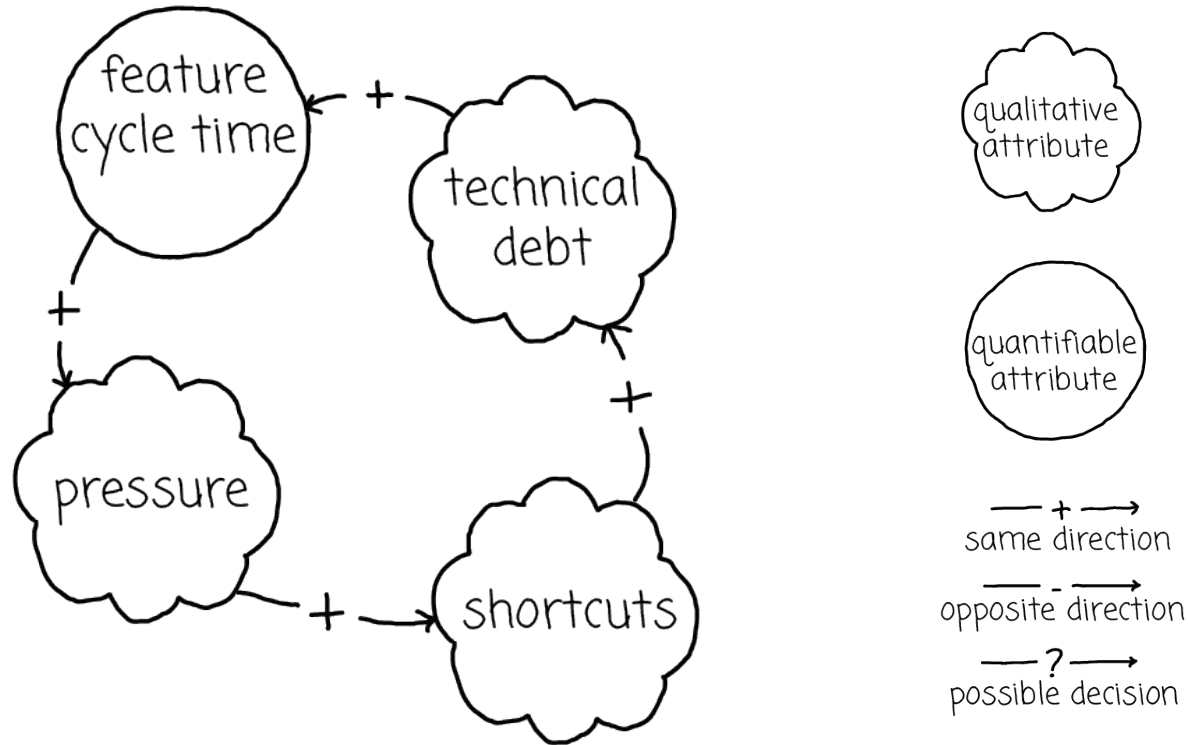
Review Boards

Freezes

Best Practices



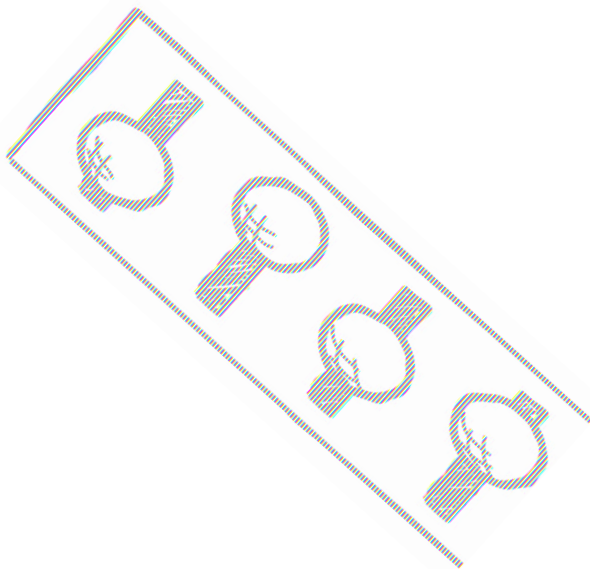
Seeing Systemic Influences with Causal Models



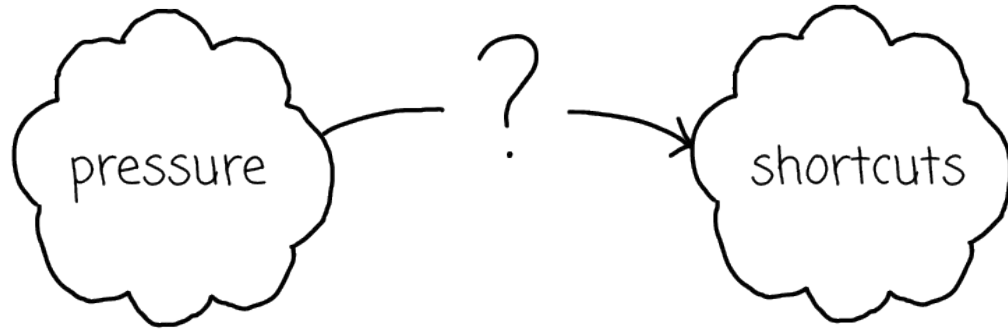
Levers

A **lever** is a choice that causes a shift in the system.

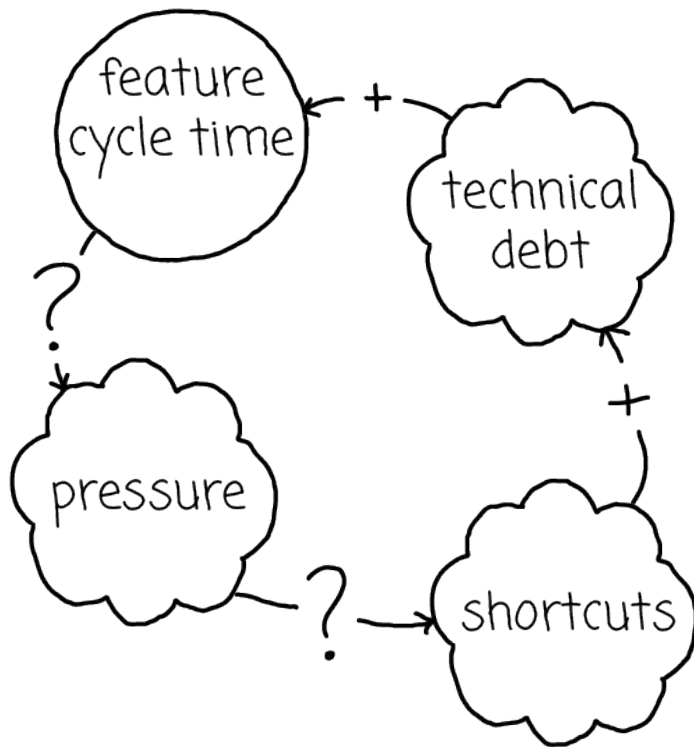
If you do something and the system does not appear to change at all, either you're not looking at the right signals, or the choice you made wasn't actually a lever.



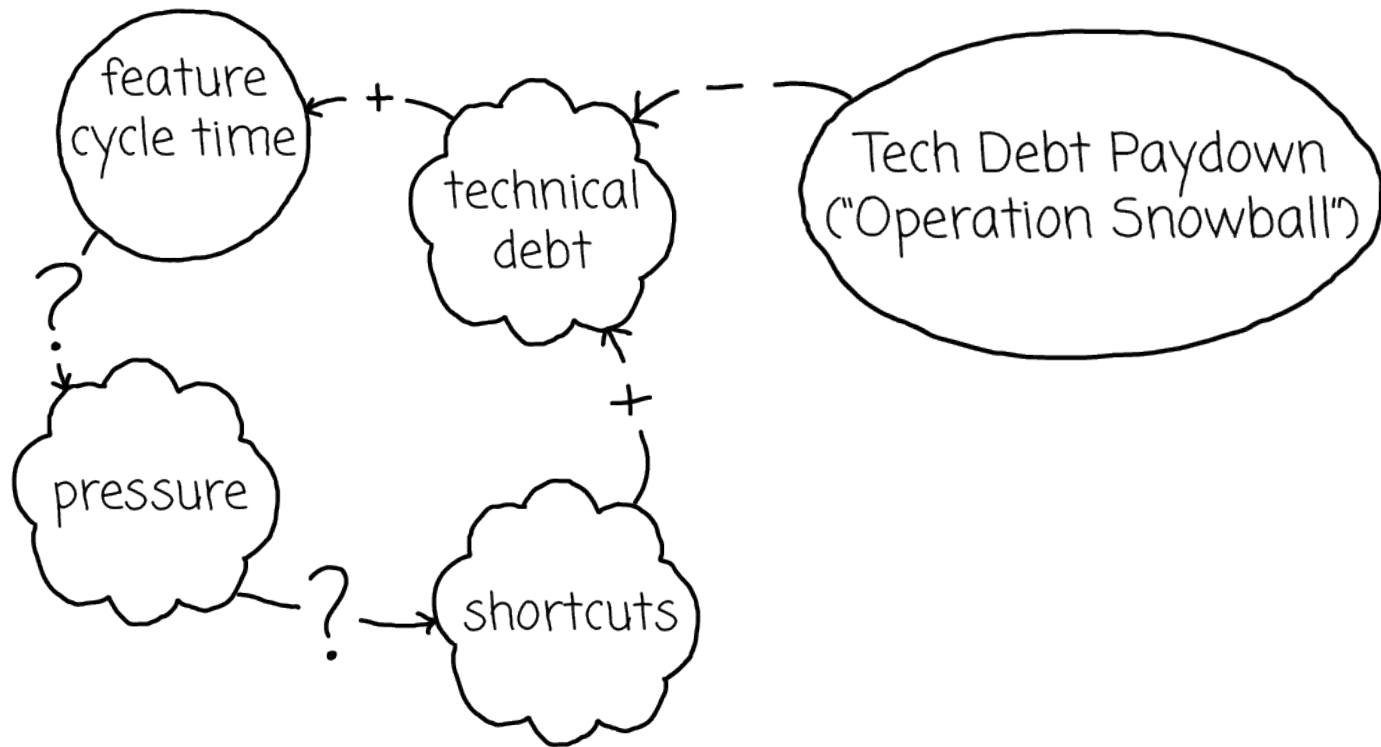
Lever 1: Identify Where You Have a Choice...



...Then Make Different Choices



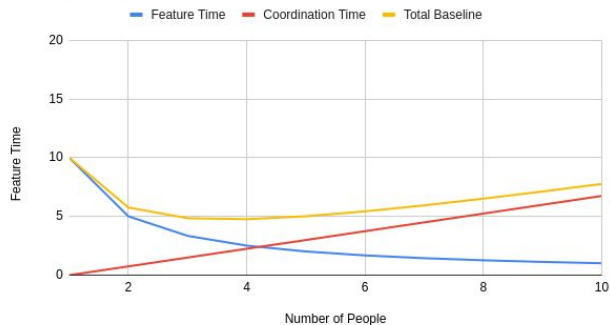
Lever 2: Add a Counterbalancing Influence



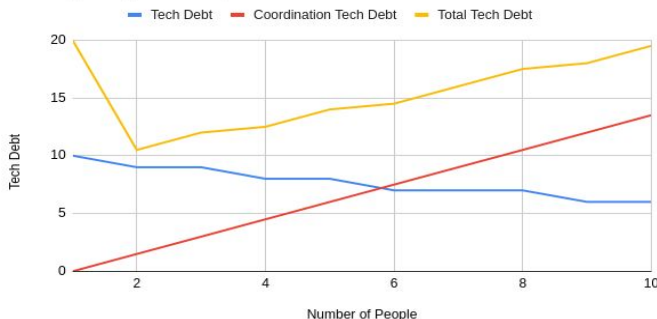
Context and Variability

See What Is Appropriate In Your Context

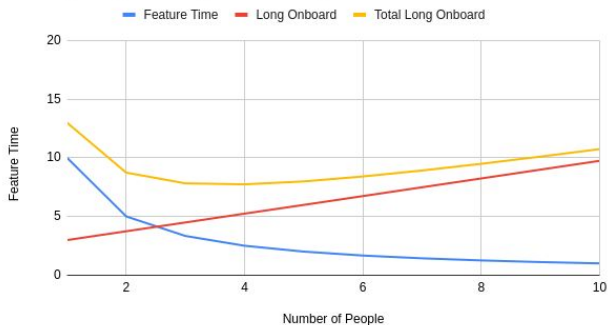
Adding People Baseline



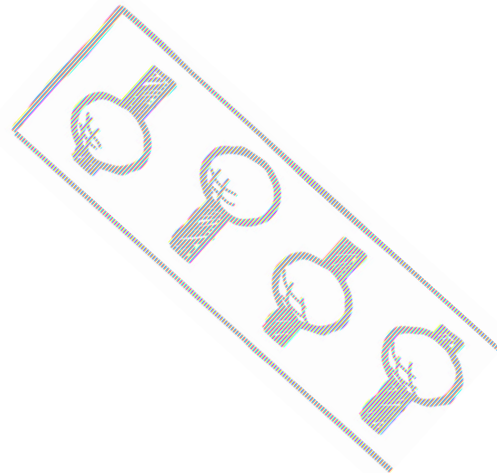
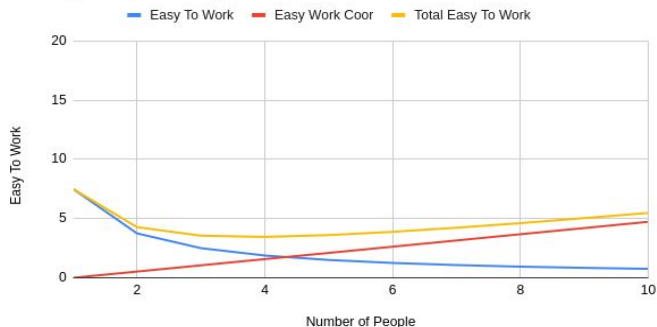
Adding People - Complex Code



Adding People - Long Onboarding Process



Adding People - Easy To Work



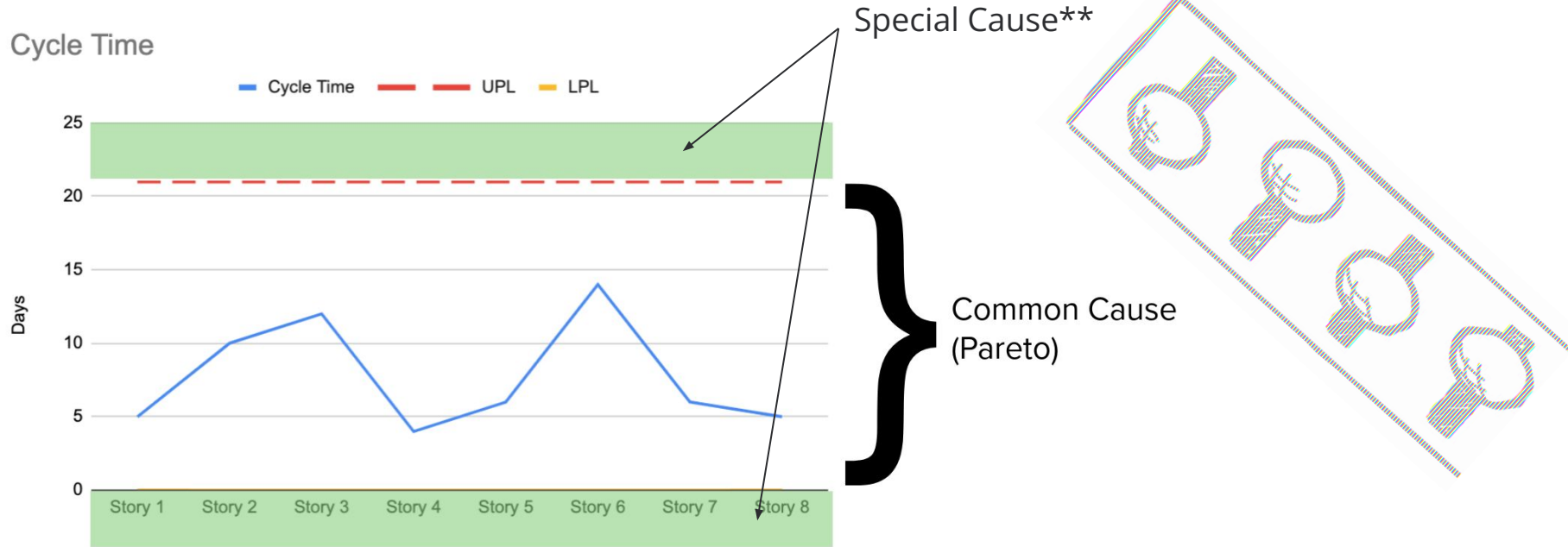
There Are No Best Practices



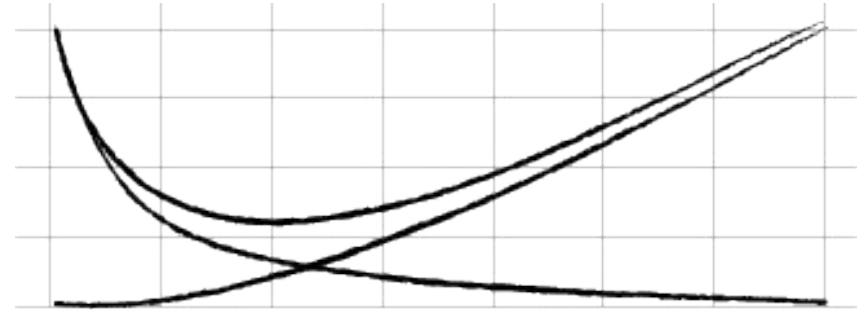
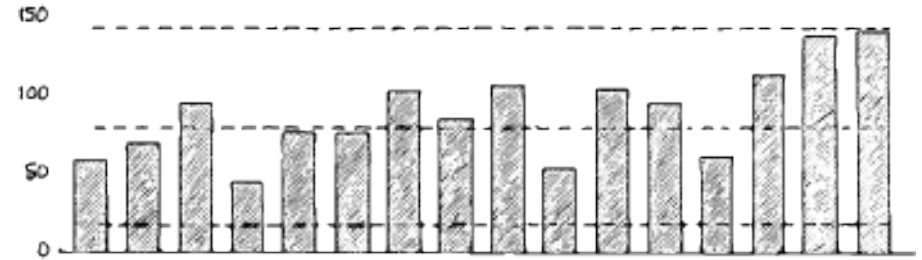
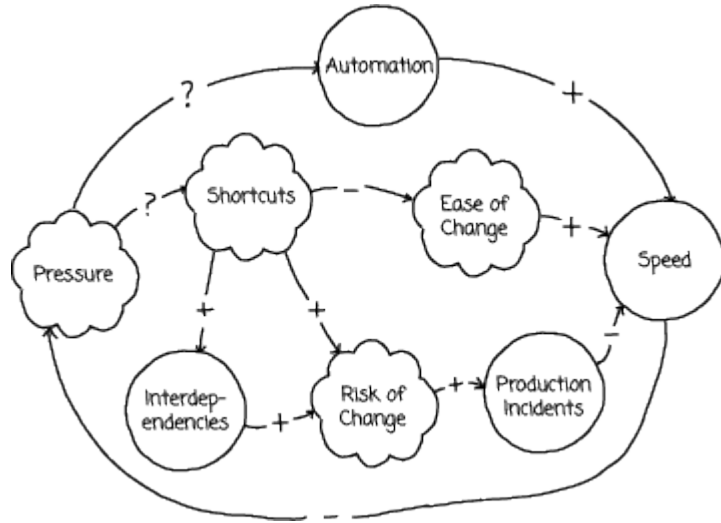
The orange line is the overall score. Above zero, Signal outweighs Noise; below zero, Noise outweighs Signal. Dashed lines show the two components. Blindspots are not yet modeled.

From - <https://sim.signalsandlevers.com/scenarios/testing-pyramid>

Avoid Tampering



Use Systems Thinking Collectively



Summary

Act where the pain is caused...

...not where it is observed.

What Questions Do You Have?

3 Illusions in Software Delivery



Thank You!

hello@signalsandlevers.com

<https://signalsandlevers.com/3IllusionsPrDC>

